



Hot Spots in Our Playground

Where is the hottest place at our school, and how could we cool it down?

Cities can get much hotter than the countryside. Some places at school feel hot and some feel cool. Let's measure them with a micro:bit thermometer!

TIME

60–90 minutes (mostly outdoors)

GROUPS

Groups of 3

AGES

Starter 6–8 · Explorer 9–11

Learning goals

- ✓ I can measure temperature with a micro:bit.
- ✓ I can do a fair test by waiting and not holding the thermometer.
- ✓ I can suggest a way to make a hot place cooler.

STEAM in this activity

S

SCIENCE

Sunlight warms surfaces; shade and light colours keep things cooler.

T

TECHNOLOGY

A micro:bit becomes a digital thermometer.

E

ENGINEERING

Plan a shade or cool roof for the hottest spot.

A

ART

Draw a colour-coded "Cool School" map.

M

MATHS

Read, compare and order temperatures in °C.



Before the lesson

- ☐ Load the thermometer program (Activity 3) onto every micro:bit and test it.
- ☐ Choose a sunny day. Walk the route and pick sunny, shady, grass, hard and sandy spots.
- ☐ Cut black and white paper, one sheet of each per group.
- ☐ Plan drinking water, hats and shade breaks.



Safety

- Hold the micro:bit by its edges.
- Keep the battery pack dry.
- Wear a hat and drink water outside.
- Stay where your teacher can see you.
- Do not touch very hot ground for long.



Kit per group

- ☐ A micro:bit with a battery pack
- ☐ The thermometer program from Activity 3
- ☐ A sunny spot and a shady spot
- ☐ Hard ground, grass and sand, all in the sun
- ☐ A sheet of black paper
- ☐ A sheet of white paper

Running order (80 minutes)

0–10 min	Story and big question. Show the micro:bit thermometer.
10–20 min	Get ready: classroom reading, practise waiting 2 minutes.
20–35 min	Test 1: sun or shade.
35–55 min	Test 2: which ground is hottest.
55–70 min	Test 3: black or white paper.
70–80 min	Share results. Where would you plant a tree?

The investigations

Start here: build: Get your thermometer ready

Children record: Temperature in the classroom

What scientists often see: A thermometer tells us how hot or cold something is. We measure temperature in degrees Celsius, written °C.

Test 1: Sun or shade?: Is it hotter in the sun or in the shade?

Children record: What does the micro:bit say?; Draw the place

What scientists often see: Sunlight warms the ground and the air. Shade blocks the sunlight, so shady places stay cooler. Trees give shade and cool cities down.

Test 2: Which ground is hottest?: Which ground gets the hottest in the sun?

Children record: What does the micro:bit say?; Draw the place

What scientists often see: Hard, dark ground like concrete and roads soaks up sunlight and gets very hot. Grass has water inside it, and water helps keep it cooler.

Test 3: Dark or light roof?: Which colour gets hotter in the sun?

Children record: What does the micro:bit say?; Draw the place

What scientists often see: Dark colours soak up more sunlight. Light colours bounce more of it away. Painting roofs white is one way cities stay cooler.

Talk about it

- Which place was hottest? Why do you think so?
- Why did we wait 2 minutes and not hold the micro:bit?
- Where at school would a tree or a white roof help most?

What to look for

Starter: Reads a number from the micro:bit and says which place was hotter.

Explorer: Explains why waiting and not holding the micro:bit makes the test fair.

Art corner

Cool School map. Draw a map of the playground. Colour hot places red and cool places blue. Draw one thing that would cool a hot place.

Maths corner

Starter: Which number is bigger: sun or shade? Circle it.

Explorer: Sun minus shade = ____ degrees. Put all your readings in order, smallest first.

Real-world link

City planners measure hot spots too. They plant trees and use light-coloured roofs to make streets cooler.

Going further

Build a shade from card or cloth over the hottest spot. Measure again. Did it work?

Curriculum links (for teacher review): NGSS K-PS3-1 (proposed) · NGSS K-PS3-2 (proposed)

WeLearn Growth Hub · Teacher guide

Activity 1: Hot Spots in Our Playground